

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 2

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 3 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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PHYS10302 Vibrations and Waves

Week 2 Questions

1. (a) A mass of $m = 0.300 \text{ kg}$ is attached to a spring of spring constant $k = 500 \text{ N/m}$. It is subject to a damping force $-(0.100 \text{ kg s}^{-1})v$, where v is the velocity. Calculate the undamped angular frequency ω_0 , the values of γ and Q , and the actual angular frequency ω .
- (b) A mass of 2.4 kg is attached to a spring of spring constant $6.0 \times 10^3 \text{ N/m}$. It has a Q value of 25. What is the damping force?
- (c) In a damped system with $Q = 123.4$, how many complete cycles does it take for an excitation to have decreased by half of its amplitude? (hint: given the amplitude falls like $\exp(-\gamma t/2)$, write this in terms of Q and then use the fact that $t = nT$, where n is the number of cycles.)

2. Consider two pendula with identical lengths and bob diameters, but with one made of aluminium with a density of $\rho_a = 2.7 \times 10^3 \text{ kg m}^{-3}$ and the other made of brass with a density of $\rho_b = 8.5 \times 10^3 \text{ kg m}^{-3}$. Write down the equations of motion of the two pendula, assuming a damping force proportional to velocity and letting b be the damping constant. What are the amplitudes of displacement of the two masses as a function of time assuming light damping? Hence explain why the rate of amplitude decrease is different for the two pendula.

Both pendula are set in motion at the same time with the same amplitude. After 10 minutes the amplitude of the aluminium pendulum has decreased to half of its initial value. What factor has the amplitude of oscillation of the brass pendulum decreased by in the same time?

3. Let's now practice some complex number algebra relevant to the mathematics of waves and vibrations.

- (a) By considering the series expansion of $\exp(i\theta)$, prove the standard formula

$$\exp(i\theta) = \cos(\theta) + i \sin(\theta).$$

- (b) By considering the real and imaginary parts of the product

$$\exp(i\theta) \exp(i\phi),$$

prove the standard sine and cosine addition formulas:

$$\cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi,$$

$$\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi.$$

- (c) Consider the differential equation

$$\frac{d^2x}{dt^2} = +\omega^2 x.$$

E&M Tutorial Sheet 2: Gauss' Law with Uniform Charge Distributions

- 1) (Revision) Write down and learn by heart the formulae for the following geometric quantities:
- (a) Circumference of a circle
 - (b) Area of a circle
 - (c) Surface area of a sphere
 - (d) Volume of a sphere
 - (e) Surface area of the curved surface of a cylinder
 - (f) Volume of a cylinder

- 2) Write down the equation describing Gauss's Law. Explain in words what each of the symbols mean. Explain in words what the equation itself means. *In an examination, if you are asked to state Gauss's Law and explain what it means then this is the kind of thing you need to be able to do.*

- 3) Sketch the *direction* of the electric field inside and outside a solid sphere of uniform charge density ρ and radius a . Make some statement about how the field varies in space. Write a few sentences to justify qualitatively the field pattern you've drawn, making references to things like symmetry and superposition.

Choose an appropriate surface and use Gauss's Law to calculate the electric field in both of these regions.

- 4) Sketch the *direction* of the electric field inside and outside an infinitely long solid cylinder of uniform charge density ρ and radius a . Make some statement about how the field varies in space. Justify it in a similar fashion to Q 3.

Choose an appropriate surface and use Gauss's Law to calculate the electric field in both of these regions.

Think carefully about the length of your Gaussian surface and check that you include the contribution to the surface integral from all parts of the surface.

- 5) Sketch the *direction* of the electric field around an infinitely large surface of uniform charge density σ . Before doing any calculation, choose an appropriate surface and use Gauss's Law quantitatively to state how the field will vary with distance from the surface.

Use Gauss's Law to calculate the electric field.

i. Show that

$$x(t) = A \exp i(\omega t + \phi)$$

is not a solution. What feature of the trial solution is causing a problem?

ii. Hence show that the general solution can be written as

$$x(t) = A_1 \exp(-\omega t) + A_2 \exp(+\omega t).$$

If the solution to the standard oscillatory differential equation can be written in terms of sines and cosines, in terms of which functions can the solutions to the differential equation in this problem be written? How do these functions behave at large t ?

PHYS10372 Mathematics 2: Tutorial-02 Questions

These questions cover the material from week-2 of PHYS10372: Lagrange Multipliers and finding div, grads and curls.

1. Evaluate $\nabla(z)$ in both Cartesian and spherical polar co-ordinates, and hence prove that:

$$\hat{\mathbf{k}} = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}$$

2. Use the method of Lagrange multipliers to find the minimum and maximum values of the function:

$$f(x, y) = x^2 + y^2$$

subject to the constraint:

$$g(x, y) = x^2 + y^2 - xy = 1$$

[Answer: $f = 2, \frac{2}{3}$]

3. $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ is an arbitrary position vector.

Find the divergence and curl of: (i) $\mathbf{r}$ and (ii) $\frac{\mathbf{r}}{r^3}$

4. Find the divergence and curl at the point (1, 2, 3) for the following vector fields:

(a) $\mathbf{v} = xy^2z\hat{\mathbf{i}} + z\hat{\mathbf{j}} + xz\hat{\mathbf{k}}$

(b) $\mathbf{v} = xe^{-y}\hat{\mathbf{i}} - ye^{-x}\hat{\mathbf{j}} + xy\hat{\mathbf{k}}$

In each case, verify that $\nabla \cdot (\nabla \times \mathbf{v}) = 0$